Elements of AIML Lab
Experiment 05 (After-submission Reference): EDA using Python

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1 How to use this reference

This reference is meant to be shared **after you submit** Experiment 05. It demonstrates a complete EDA flow and, more importantly, shows **how to write observations** (EDA is not only plots).

2 EDA in one sentence

EDA answers: *"What does this dataset look like, what can go wrong, and what should we do next?"*

3 Recommended EDA sequence (teaching order)

3.1 1) Load + understand structure

```
import pandas as pd

df = pd.read_csv("processed.csv") # use Exp-04 output if available
```

```
print(df.shape)
print(df.head())
print(df.info())
```

What to write: mention rows/columns and the main column types (numeric/categorical/date).

3.2 2) Data quality checks (missing + duplicates)

```
miss = df.isna().sum().sort_values(ascending=False)
print(miss[miss > 0])
print("Duplicates:", df.duplicated().sum())
```

What to write: name the top 2–3 columns with missing values and their counts/percentages.

3.3 3) Descriptive summaries

```
print(df.describe())
print(df.describe(include="object"))
```

What to look for: min/max surprises, skewed distributions, rare categories.

3.4 4) Univariate plots (distribution of one feature)

```
import matplotlib.pyplot as plt
import seaborn as sns

sns.histplot(df["num_col"], kde=True)
plt.title("Distribution of num_col")
plt.show()

sns.boxplot(x=df["num_col"])
plt.title("Boxplot of num_col")
plt.show()
```

Observation examples:

- "`num_col` is right-skewed; most values are below 50 with a long tail up to 300."
- "The boxplot shows extreme values; we may need outlier handling or log transform."

3.5 5) Relationship plots (feature vs feature)

```
corr = df.select_dtypes("number").corr()
sns.heatmap(corr, cmap="coolwarm", center=0)
plt.title("Correlation heatmap (numeric features)")
plt.show()
```

How to interpret:

- strong positive correlation $\approx$ features increase together,
- strong negative correlation $\approx$ one increases as the other decreases,
- near zero correlation does not guarantee independence (could be non-linear).

3.6 6) If there is a target column (optional but valuable)

- Numeric target: scatter plot feature vs target.
- Class label: compare distributions by class (boxplots or histograms per class).

4 Minimum plot set (good standard)

- One histogram (distribution) + written comment.
- One boxplot (outliers/spread) + written comment.
- One relationship plot (scatter or heatmap) + written comment.

5 Common EDA writing pattern (use this in reports)

For each plot/check, write:

1. **Evidence:** what you saw (numbers/plot pattern).
2. **Meaning:** what it suggests about the data.
3. **Action:** what you would do next (missing handling, scaling, encoding, etc.).

6 Troubleshooting

- **Plots look empty:** check column names, data types, and missing values.
- **Too many categories:** plot top-k value counts and group rare categories as "Other".
- **Time columns:** convert with `pd.to_datetime` before time-based plots.

Sources: Provost and Fawcett, *Data Science for Business.*;McKinney, *Python for Data Analysis.*;WEKA Documentation (Waikato Environment for Knowledge Analysis), accessed 2026-02-28.;Matplotlib and Seaborn documentation, accessed 2026-02-28.